

Farnaz Alam Ahmed

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Education

Carnegie Mellon University, Pittsburgh, PA

May 2027

Master of Science in Robotic Systems Development

GPA: 4.08/4

Relevant Coursework: Computer Vision, Robot Mobility & Autonomy, Deep Learning, System Engineering

National Institute of Technology, Silchar, India

May 2022

Bachelor of Technology in Mechanical Engineering

CGPA: 9.5/10

Skills

Programming & Software: Python, C++, C, Matlab, PyTorch, OpenCV, Linux, Git

Robotics & Simulation: ROS2, MuJoCo, Isaac Sim, Isaac Lab, COLMAP, Gazebo, Unity

Hardware & Design: Docker, CAD Modelling, NVIDIA Jetson, sensor integration, Military-Grade Qualification Testing, JSS 55555 Environmental Testing

Work Experience

AI Intern, Agility Robotics, Bangalore

Jun 2026 - Present

- Building a Universal Manipulation Interface gripper-to-policy pipeline for humanoid Digit, converting multimodal MCAP demonstrations into synchronized datasets used to train, evaluate, and improve learned manipulation policies.
- Developing flow-matching-based policies for Digit and using multimodal robot logs to refine policy behavior, trajectory quality, and sensor-action alignment.

R&D Engineer (Radar Systems), Bharat Electronics Ltd., Bangalore

Oct 2022 - Jul 2025

- Led end-to-end design, qualification, and integration of radar subsystems (Active Array Antenna Unit, Interrogator, and Exciter Receiver Unit), now operational in production at multiple defense sites.
- Developed simulation and visualization environments in Unity to support operator training.

Research Assistant, Illumination and Imaging (ILIM) Laboratory, CMU

Nov 2025 - Apr 2026

- Built a 3D reconstruction pipeline using COLMAP for the ULTRRA Challenge dataset, achieving 98.75% image registration, generating 12.9M+ point clouds, and extracting intrinsic/extrinsic parameters for 300+ cameras.
- Evaluated 3D reconstruction methods (VGGT, MapAnything) vs. COLMAP, achieving $\sim 2.6\times$ lower reconstruction error ($1.741 \rightarrow 0.382$) and improving point cloud accuracy to 0.677 with bundle adjustment.

Technical Assistant, JP Morgan Chase & Co. AI Maker Space [Project]

Aug 2025 - Jan 2026

- Fine-tuned Isaac GR00T (N1.6-3B) on Kinova Gen3 pick-and-place demonstrations and deployed a client/server control interface for real-time execution on the robot.

Projects

Deep Skill Graph for Humanoid Navigation in Cluttered Space [Project]

Jan 2026 - May 2026

- Trained a PPO-based low-level locomotion policy and deployed it across Sim2Sim and Sim2Real settings.
- Developed Transformer- and MLP-based transition dynamics model and integrated with an MPC planner for model-based reasoning, achieving up to $3\times$ reduction in rollout error.
- Trained a global policy using TD3 and leveraged it to iteratively learn and expand skill primitives within the graph, achieving +35% improvement in navigation success rate.

Automated Lab, MRSD Capstone Project [Project]

Aug 2025 - Present

- Built an end-to-end perception pipeline with multi-stage (overhead + wrist camera) sensing for coarse-to-fine localization and precise 3D pose estimation.
- Deployed real-time detection and segmentation models (YOLO v11) on NVIDIA Jetson Xavier, enabling distributed edge processing via WebSocket-based image and point cloud streaming.
- Developing an anomaly detection pipeline in lab set-ups using CLIP ViT-B/32 frame embeddings, and a dual-transformer spatio-temporal model.

Publications

Hrishav Dey, Sumit Kumar Mehta, Farnaz Alam Ahmed, et al. "Numerical investigation of thermo-hydraulic features of viscoplastic flow in wavy channels."